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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL 2021

CH305S – INORGANIC CHEMISTRY – II

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Among the following,which can be used to estimate Al^{3+}.
   a) DMG b) Alizarin c) Magneson d) Cupferon

2. Aprotic solvent is _____.
   a) Ammonia b) Pyridine c) HCl d) Benzene

3. The element shows maximum tendency for catenation is ,
   a) Carbon b) Silicon c) Lead d) Germanium

4. Solid carbon dioxide is known as _____.
   a) Water glass b) Dry ice c) Heavy water d) Soft water

5. Caro's acid is _____.
   a) H_2SO_3 b) H_2SO_4 c) H_2SO_5 d) $H_2S_2O_8$

6. Which metal is passive made by pure nitric acid?
   a) Zn b) Cu c) Ag d) Fe

7. Select the most stable compound from the following
   a) IF b) BrF c) ClF d) ICl

8. Clathrates are formed only by _____ group elements in the periodic table.
   a) Zero b) First c) Second d) Third

9. Heating the ores strongly in the presence of air is called _____
   a) Roasting b) Quenching c) Calcination d) Annealing

10. Carnitite is the ore of
    a) Ti b) V c) W d) Cr

II. Fill up the blanks

11. Thiourea reagent is used to detection of _____.

12. Silicon is a _____ conductor.

13. When hydroxyl amine react with carbonyl compounds gives _____.

14. CN^- is used as a _____.

15. _____ element is used as anticathode in x-ray tubes.

III. Match the following

16. NH_4^+	a. Paramagnetic
17. Ortho silicates	b. Froth floatation
18. Oxygen	c. Nesseler's reagent
19. XeF_4	d. Wollastonite
20. Sulphur ores	e. Square planar

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. Explain solubility product.

22. What are aqueous and non-aqueous solvents?

23. CO_2 is a gas whereas SiO_2 is a solid- Account for.

24. Write a short note on 3D silicates.

25. What is the action of hydrazine with, i) Ozone and ii) Cupric salts

26. What is the oxidation number of sulphur in two peracids of sulphur?

27. Write any four unique properties of fluorine.

28. Write a short note on clatharates.

29. How impure vanadium is purified?

30. Explain the process of magnetic separation method.

31. Calculate the oxidation number of Mn in $KMnO_4$.

32. Write a short note on amphiboles.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) Explain with suitable examples of ammonolysis reactions. **(2)**
    b) What are advantage and disadvantages using HF as a solvent? **(4)**
    c) Draw the structure of DMG. **(1)**

34. a) Bring out the similarities and difference in Carbon and Silicon compounds. **(4)**
    b) Balance the following equation by using ion electron method. **(3)**
    $Cr_2O_7^{2-} + Fe^{2+} + H^+ \rightarrow Cr^{2+} + Fe^{3+} + H_2O$.

35. a) Write down the properties of NH_3 and PH_3 with reference to ,
i) Thermal stability and ii) Basic character. **(4)**
b) Compare Caro's and Marshall's acids. **(3)**

36. a) Tabulate the hybridization and geometry of any one example from the types of AX, AX_3, AX_5 and AX_7 in inter halogen compounds. **(4)**
b) Discuss the position of inert gases in the periodic table. **(3)**

37. a) Explain the extraction of titanium from its ore. **(4)**
b) Why transition metal ions show colors? Explain with suitable examples. **(3)**

38. a) Calculate the oxidation number for the underlined elements in the followings. **(4)**
a) $\underline{\mathbf{N}}H_2OH$ b) $\underline{\mathbf{Pb}}_3O_4$ c) $Na\underline{\mathbf{Cr}}O_2$ d) $H_2S_2O_7$
b) Discuss the preparation, properties and uses of calcium carbide. **(3)**

39. a) Cupferon is used to identify the _____element. **(1)**
b) Carbon has not an unique quality of forming multiple bonds.(True/False) **(1)**
c) Complete the equation $AgNO_3 + N_2H_4 \rightarrow$ _____. **(1)**
d) Mellitic acid is one of the oxidation products of _____. **(1)**
e) The anhydride of nitrous acid is _____. **(1)**
f) The hybridized state of X in XO_3^- is _____. **(1)**
g) _____ is an example for tetrathionic acid. **(1)**

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